Premium Auto Services: Smart Vehicle Management System

Comprehensive Project Report

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1. Executive Summary

The **Premium Auto Services - Smart Vehicle Management System** is an enterprise-grade, full-stack software solution designed to revolutionize modern garage and auto-repair shop management. Combining a robust Java Spring Boot backend, an interactive React frontend, and an advanced AI-powered Automatic License Plate Recognition (ALPR) Python engine, the system automates vehicle entry, inventory tracking, service scheduling, employee registry, and financial reporting.

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2. Market Differentiation & User Benefits

Why Users Will Use This Application

Traditional garage management software relies heavily on manual data entry, fragmented Excel sheets, and disconnected systems for billing and inventory. This system offers a unified, automated ecosystem.

What is Different Than Competitions

- **AI ALPR Gate Integration:** Competitors require manual check-ins. This system uses a custom AI engine (OpenCV + EasyOCR) to automatically log arriving vehicles, recognize Indian HSRP plates, apply fuzzy database matching, and check them against registered records in milliseconds.
- **Automated Inventory Restocking:** The "Parts Vault" inherently tracks low stock and triggers automated vendor orders the moment inventory drops below threshold.
- **Predictive Maintenance:** Leveraging odometer readings to dynamically calculate and alert users of upcoming service milestones.
- **Role-Based Access Control (RBAC):** Built-in security layers tailored specifically to auto-shops (Owner vs. Manager vs. Employee).

Benefits to the User

- **Time Savings:** Automated gate logs and one-click invoice PDF generation.
- **Cost Efficiency:** Strict inventory tracking prevents part pilferage and out-of-stock scenarios.
- **Business Insights:** Real-time dashboards showing gross revenue, daily service volumes, and ALPR traffic.

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3. Top Features & Feature Uses

1. **AI Gate Scanner (ALPR):** Automatically logs vehicle arrivals at the facility gate.
2. **Service Bay (Maintenance):** End-to-end ticket management from vehicle check-in to mechanic assignment, tracking aftermarket mods, and PDF invoice generation.
3. **Parts Vault (Inventory):** Tracks parts, prices, locations, and handles automated restocking rules with vendors.
4. **Registry & Access Control:** Manages Employees (including document uploads via BLOBs), Vendors, and System Users.
5. **SmartHealth & Alerts:** Predictive insights for vehicle maintenance based on mileage.
6. **Service Reports:** Advanced multi-filter financial and operational reporting.

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4. Workflow & Page-by-Page Breakdown

- **Overview:** The central dashboard. Displays aggregate metrics (Total Vehicles, ALPR Scans, Active Services), ALPR traffic charts, and Today's Scheduled Services.
- **Vehicles:** A searchable, filterable directory of all registered vehicles (Car, Motorbike, Heavy Vehicle, etc.). Allows editing owner details and vehicle metrics.
- **Maintenance (Service Bay):** The core operational page. Service advisors create tickets, assign registered mechanics, quote prices, and track status. Once complete, tickets move to Reports.
- **Reports:** Historical ledger of all completed services. Features advanced filtering (Date, Payment Status, Vehicle Plate) and ALPR history logs. Allows generating PDF invoices.
- **Parts Vault:** Inventory management. Employees log new parts. Owners can define "Automated Order" rules to instantly buy parts from specific vendors when stock is low.
- **Registry:** Human Resources and IT administration. Registers employees (mechanic, manager, cashier, etc.) and stores KYC documents (Aadhar, PAN, Resume). Manages RBAC User credentials and Vendors.
- **GateScanner:** Interface for the AI Python Sidecar. Connects to webcams or manual uploads to run the OpenCV/EasyOCR ALPR model.

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5. System Architecture & Diagrams

The architecture follows a Microservices-inspired Monolithic layout.

- **Frontend (React):** Single Page Application handling UI and state.
- **Backend (Spring Boot):** Core business logic, JPA ORM, and JWT security.
- **AI Sidecar (Python/FastAPI):** Dedicated image processing engine.

```mermaid
flowchart TD
    subgraph Frontend [React SPA]
        UI[User Interface]
        API_CLIENT[Axios / Fetch Client]
    end

    subgraph Backend [Java Spring Boot]
        SEC[Spring Security / JWT]
        CTRL[REST Controllers]
        SERV[Business Services]
        REPO[JPA Repositories]
    end

    subgraph AI_Engine [Python FastAPI]
        CV[OpenCV Preprocessing]
        OCR[EasyOCR Engine]
        LOGIC[Indian HSRP Formatting]
    end

    subgraph Database [MySQL]
        DB[(Relational Tables & BLOBs)]
    end

    UI -->|HTTPS Request| API_CLIENT
    API_CLIENT -->|REST API + JWT| SEC
    SEC --> CTRL
    CTRL --> SERV
    SERV <--> REPO
    REPO <--> DB

    CTRL -.->|Image Bytes (Port 8000)| AI_Engine
    AI_Engine -.->|License Plate String| CTRL
    CTRL -->|Levenshtein Fuzzy Match| DB
```

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6. Technology Stack & Justifications

- **Frontend: React.js (HTML5/CSS3/JS)**
    - *Reason:* Component-based architecture allows for highly dynamic, responsive dashboards without page reloads.
- **Backend: Java Spring Boot**
    - *Reason:* Enterprise-grade stability, excellent ORM (Hibernate) support for complex relational schemas, and robust built-in security frameworks.
- **AI Engine: Python (FastAPI, OpenCV, EasyOCR, NumPy)**
    - *Reason:* Python is the undisputed industry standard for Computer Vision. FastAPI allows for lightning-fast, non-blocking asynchronous image processing.
- **Database: MySQL**
    - *Reason:* ACID compliance is mandatory for financial invoices, employee records, and inventory tracking.
- **Libraries:** `jsPDF` (Invoice Generation), `Recharts` (Dashboard charts).

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7. Backend, Database, and API Documentation

Database Schema Core Entities

- **User:** `id`, `username`, `password` (BCrypt), `role` (Owner/Manager/Employee).
- **Vehicle:** `id`, `licensePlate`, `makeAndModel`, `ownerName`, `type`.
- **ServiceTicket:** `id`, `vehicle_id`, `mechanic_id`, `status`, `paymentStatus`, `estimatedCost`.
- **InventoryItem:** `id`, `partName`, `stockLevel`, `autoOrderVendorId`.
- **Employee:** `id`, `name`, `post`, `salary`, `aadharCard` (LONGBLOB), `photo` (LONGBLOB).

Core REST APIs

- `POST /api/auth/login`: Authenticates and returns JWT.
- `GET /api/services`: Retrieves service queue.
- `PUT /api/services/{id}/paymentStatus`: Updates financial ledger.
- `POST /api/registry/employees`: Accepts `multipart/form-data` for employee KYC document uploads.
- `POST /api/alpr/scan`: Relays base64 images to Python, validates with Levenshtein fuzzy match, and saves to `GateScanLog`.

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8. Security Architecture

- **Authentication:** Stateless JSON Web Tokens (JWT) eliminate session hacking and scale perfectly.
- **Password Protection:** BCrypt hashing ensures passwords are mathematically irreversible in the database.
- **Role-Based Access Control (RBAC):**
    - `Owner`: God-mode. Can delete users, delete services, and view all data.
    - `Manager`: Can manage employees and services, but cannot create/delete `Owner` accounts or wipe historical ledgers.
    - `Employee`: Limited to read/write operational data (Parts, Vehicles) with no access to HR or user management.
- **Data Protection:** ALPR manual overrides and fuzzy matching prevent SQL injection by stripping non-alphanumeric characters instantly.

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9. Future Scope & Future-Readiness

This platform is architected natively for scale. Because the AI is decoupled into a sidecar, and the backend is purely RESTful, the system is highly future-ready.

Next Possible Developments

1. **YOLOv10 Plate Detection:** Upgrading the ALPR to use a dedicated Object Detection neural network prior to OCR, achieving 99.9% accuracy in blizzard/monsoon conditions.
2. **Cloud Telematics Integration:** Expanding the "SmartHealth" module to accept live OBD-II port data via IoT protocols (MQTT) from connected cars.
3. **Mobile App Extension:** Wrapping the React frontend in React Native for customers to track their car's service status live on their phones.
4. **Real-ESRGAN Upscaling:** Implementing generative AI to hallucinate missing pixels on extremely blurry security cam footage.

Conclusion:
Premium Auto Services is not just a digital ledger; it is an intelligent, automated operational nervous system built to future-proof auto-repair businesses for the AI era.